```
    return {child.centroid for child in
this.children}                          #4
```

#1 Method `getEntriesCentroids` is defined on `SsNode`. It returns the centroids of the entries within the node's bounding envelope.
#2/#3 If the node is a leaf, we can just return its points.
#4 Otherwise, we need to return a list of all the centroids of this node's children: we use a construct typically called list-comprehension to denote this list (see appendix A).

Hang tight, we aren't done yet: I know we have been going on with this section for a while now, but we still miss one piece of the puzzle to finish our implementation of the `insert` method: the `splitPoints` helper function.

This method might seem trivial, but it's actually a bit tricky to get it right: let's say it needs at least some thought.

So, let's first go though an example, and then write some (pseudo)code for it! Figure 10.18 illustrates the steps we need to perform such a split. We start with a node containing 8 points. We don't know, and don't need to know, if those are dataset points or nodes' centroids: it is irrelevant for this method.

Suppose we have computed the direction of maximum variance, and that it is along the y axis; then we need to project the points along this axis, which is equivalent to only consider the y coordinate of the points (because of the definition of our coordinate system).